Supplementary Material: Conchordal: Emergent Harmony via Direct Cognitive Coupling in a Psychoacoustic Landscape

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S1. Algorithm Pseudocode

S1.1 Pitch Adaptation (Experiment 1)

Input: agents $\{a_1, \ldots, a_N\}$ with pitches x_i in $\log_2$-freq space.
Parameters: step size $\lambda = 0.15$ st, repulsion kernel $\sigma_r = 0.06$ st, $T_0 = 0.05$, $\tau = 30$.

`for` round $t = 1$ `to` 100:
  $T_t \leftarrow T_0 \cdot \exp(-t/\tau)$
  `for` each agent a_i:
    $\delta \sim \mathrm{Uniform}(-\lambda, +\lambda)$; $x' \leftarrow x_i + \delta$
    $\mathrm{Rep}(x) = \sum_{j \neq i} \exp\!\big(-(x - x_j)^2 / 2\sigma_r^2\big)$
    $S_{\text{old}} \leftarrow s(t) \cdot C_{\text{field}}(x_i) - \mathrm{Rep}(x_i)$
    $S_{\text{new}} \leftarrow s(t) \cdot C_{\text{field}}(x') - \mathrm{Rep}(x')$
    `if` $U(0,1) < \min\!\big(1, \exp\!\big((S_{\text{new}} - S_{\text{old}})/T_t\big)\big)$ `then` $x_i \leftarrow x'$

where $s(t) = -1$ for $t \leq 50$ (dissonance-seeking), $+1$ for $t > 50$ (consonance-seeking).

S1.2 Metabolic Policy (Experiment 2)

Parameters: $c_b = 0.5$/s, $c_a = 0.02$, $r_E = 0.5$, $\Delta t = 512/48000$ s.

`for` each time step:
  `for` each agent a_i:
    $E_i \leftarrow E_i - c_b \cdot \Delta t$ *(basal drain)*
    `if` phase ϕ_i crosses zero `and` $E_i > 0$: *(attack event)*
      $C_{\text{level01}} \leftarrow \sigma\!\big(2 \cdot C_{\text{field}}(x_i)\big)$
      $E_i \leftarrow E_i - c_a + r_E \cdot C_{\text{level01}}$
    `if` $E_i \leq 0$: agent dies; respawn at random pitch within ± 1 oct

S1.3 Kuramoto Phase Coupling (Experiment 3)

Parameters: $K_{\text{base}} = 3.0$, $E_{\text{cap}} = 1.0$, kick at 2 Hz.

`for` each time step dt:
  `for` each agent a_i:
    $v_i \leftarrow \sqrt{E_i / E_{\text{cap}}}$ *(vitality)*
    $k_{\text{eff},i} \leftarrow K_{\text{base}} \cdot v_i$
    $\dot{\phi}_i \leftarrow \omega_i + (k_{\text{eff},i}/N) \sum_j \sin(\phi_j - \phi_i) + F(t)$
    $\phi_i \leftarrow \phi_i + \dot{\phi}_i \cdot dt$
  $R(t) = (1/N) \big| \sum_j e^{i\phi_j} \big|$ *(order parameter)*

Continuous energy model: $dE_i/dt = -c_b + r_E \cdot C_i$ ($c_b = 0.05$/s, $r_E = 0.1$/s).

S1.4 Hereditary Respawn (Experiment 4)

Parameters: mutation $\sigma = 0.003\ \log_2$, repulsion $\sigma_r = 0.06$ st.

`for` each time step:
  Run metabolic policy (S1.2) with recharge enabled
  `if` agent a_i dies ($E_i \leq 0$):
    Select parent a_p uniformly from surviving agents
    $x_{\text{new}} \sim \mathcal{N}(x_p, \sigma^2)$; clamp to anchor ± 1 oct
    Respawn at x_{new} with $E = 1.0$

Random control: respawn at $\mathrm{Uniform}(\text{anchor} \pm 1\,\text{oct})$ instead.

S1b. Normalization of H_{01} and R_{01}

Both harmonicity and roughness are normalized to $[0, 1]$ before entering the consonance core.

Harmonicity (H_{01})

The raw harmonicity potential from sibling projection (Eqs. 1–2 in the main text) is normalized by dividing by the self-harmonicity of a single harmonic tone at the reference position (which equals the kernel's maximum response):

$$H_{01}(x) = \mathrm{clamp}\left(\frac{H_{\text{raw}}(x)}{H_{\text{ref,max}}},\ 0,\ 1\right), \tag{S1}$$

where $H_{\text{ref,max}} = 1.0$ (the kernel is pre-normalized so that a single harmonic stack produces unit peak).

Roughness (R_{01})

Raw roughness is first expressed as a ratio $\rho = R_{\text{raw}}/R_{\text{ref}}$, where R_{ref} is the roughness of a single-tone reference density (computed once at initialization). This ratio is then mapped through a piecewise saturation function with physi-

ological constant k ($k = 1.0$ in all experiments):

$$R_{01}(\rho) = \begin{cases} 0 & \rho \le 0, \\ \dfrac{\rho}{1+k} & 0 < \rho < 1, \\ 1 - \dfrac{k}{\rho + k} & \rho \ge 1, \end{cases} \tag{S2}$$

clamped to $[0, 1]$. For $\rho < 1$ (below-reference roughness) the mapping is linear; for $\rho \ge 1$ it is a hyperbolic saturation that approaches 1 asymptotically as $\rho \to \infty$. At $\rho = 1$ the function is continuous: $R_{01}(1) = 1/(1 + k)$. With $k = 1$, the reference roughness maps to $R_{01} = 0.5$.

S2. Complete Parameter Table

Table S1 lists all parameters used across experiments.

S3. Experiment 4: Full Sweep Results

Table S2 reports the hereditary adaptation parameter sweep. Each of the 15 conditions was run with 20 seeds per arm (heredity vs. random control). The reported setting in the main text ($\sigma = 0.003$, deaths $= 4{,}000$, $N = 32$) is marked with $\star$.

S4. Bilinear Consonance Core: Derivation

The consonance core (Eq. 4 in the main text) is the most general bilinear form for two inputs $H_{01}, R_{01} \in [0, 1]$:

$$C = a\, H_{01} + b\, R_{01} + c\, H_{01}\, R_{01} + d. \tag{S3}$$

$C_{\textbf{field}}$ $(a, b, c, d) = (1, -1.35, 1, 0)$:

$$C_{\text{field}} = H_{01} - 1.35\, R_{01} + H_{01}\, R_{01} \tag{S4}$$
$$= H_{01}(1 + R_{01}) - 1.35\, R_{01}. \tag{S5}$$

When H_{01} is high (strong harmonicity), the $H_{01}\, R_{01}$ term partially compensates for roughness, reflecting the perceptual observation that harmonically rich intervals tolerate moderate beating. When $H_{01} = 0$, roughness is penalized at full strength.

$C_{\textbf{density}}$ $(a, b, c, d) = (1, 0, -1, 0)$:

$$C_{\text{density}} = H_{01}(1 - R_{01}). \tag{S6}$$

This product is nonnegative and peaks where harmonicity is high and roughness is low. In Conchordal's interactive mode it serves as a spawn-position weight; it is not used in the experiments reported in this paper but is shown in Figure 1D for comparison.

$C_{\textbf{level01}}$: C_{field} passed through a logistic sigmoid $\sigma(\beta(x - \theta))$ with $\beta = 2, \theta = 0$. This bounds the metabolic signal to $[0, 1]$ while preserving the ranking.

S5. Sign-Flip Permutation Test

The interaction contrast for the integration test (Figure 7 in the main text) is tested with a one-sample sign-flip permutation test:

1. For each of $n = 20$ matched seeds, compute the contrast $d_i = \text{HH}_i - \text{H}_i - \text{RH}_i + \text{R}_i$.
2. Under H_0: $E[d_i] = 0$, each d_i is equally likely to be positive or negative. For each of $2^{20} = 1{,}048{,}576$ sign patterns $s \in \{-1, +1\}^{20}$, compute $\bar{d}_s = \frac{1}{n} \sum_i s_i\, d_i$.
3. The p-value is the fraction of sign patterns yielding $|\bar{d}_s| \ge |\bar{d}_{\text{obs}}|$.

Because $n = 20$, exact enumeration is feasible. The observed $p = 0.000053$ (1 in 18,868 patterns) with mean contrast 0.30 (95% CI $[0.20, 0.39]$).

S6. Consonance-Only Control (Experiment 1)

Experiment 1 uses a two-phase curriculum: dissonance-seeking (rounds 1–50) followed by consonance-seeking (rounds 51–100). To assess whether the resulting polyphony depends on this schedule, we ran a *consonance-only* control with identical parameters but $s(t) = +1$ for all 100 rounds (baseline condition, 20 seeds).

The consonance-only condition still produces structured polyphony (14.8 unique pitch bins, well above the collapse level of $\approx$13 observed without repulsion in the main text) and achieves *higher* mean consonance ($C_{\text{score}} = 0.54$ vs. 0.47), confirming that consonant structure emerges from the landscape–repulsion interaction alone. However, the curriculum significantly improves diversity: more unique pitch bins (19.2 vs. 14.8), wider nearest-neighbour spacing (0.57 vs. 0.35 st), and higher interval entropy (3.93 vs. 3.53 nats). The dissonance-seeking phase thus serves as an initialisation strategy that broadens the starting distribution, but the self-organised polyphony reported in the main text is not an artefact of the curriculum schedule.

S7. Terrain Control and Robustness Analysis

Three supplementary analyses test whether the self-organised polyphony reported in Experiment 1 depends on the specific structure of the psychoacoustic landscape rather than on incidental parameter choices.

S7.1 Shuffled-Landscape Control

We destroy spatial coherence by applying a fixed random permutation of the C_{score} bin indices, generated once per run and held fixed throughout all sweeps. Agents therefore move on a stationary landscape with the same *marginal* score distribution but no local gradient structure. Table S4 compares the shuffled condition with the standard baseline ($n = 20$ seeds each).

C_{score} (ref) denotes the consonance score of each final configuration evaluated on the *original* (unshuffled) landscape, enabling a fair cross-condition comparison. Without coherent spatial gradients, agents still spread across the register (similar unique-bin count and nearest-neighbour spacing) thanks to the repulsion term, but the interval structure degrades sharply: entropy rises from 3.93 to 4.65 nats ($p < 0.0001$), approaching the uniform distribution. The coherent psychoacoustic landscape is therefore necessary for agents to self-organise into musically structured pitch configurations.

S7.2 Harmonicity–Roughness Misalignment Control

The consonance kernel combines harmonicity (H_{01}) and roughness (R_{01}). We test whether their spatial alignment matters by circularly shifting R_{01} by Δ cents before scoring, which preserves each component's marginal statistics but breaks the psychoacoustically motivated registration between them. Four shift values span the range $\Delta \in \{137, 271, 389, 523\}$ cents. Table S5 reports the results ($n = 20$ seeds per shift, 80 runs total for the misaligned aggregate).

No individual shift or the aggregate reaches significance on any terrain-agnostic metric (all $p > 0.17$). This suggests that at the coarse resolution of the E2 simulation (24 agents, semitone-scale steps), moderate H–R misalignment does not destroy polyphonic structure. The shuffled-landscape control (S7.1) remains the stronger test, demonstrating that *some* form of coherent landscape gradient is essential.

S7.3 Consonance Kernel Coefficient Sensitivity

The consonance kernel $C = a\,H_{01} + b\,R_{01} + c\,H_{01}\,R_{01} + d$ has four coefficients; the default is $(a, b, c, d) = (1.0, -1.35, 1.0, 0.0)$. We sweep $b \in \{-2.0, -1.35, -0.7, 0.0\}$ and $c \in \{0.0, 0.5, 1.0, 1.5\}$ with $a = 1$, $d = 0$ fixed (16 conditions, 20 seeds each). Table S6 shows all conditions produce unique-bin counts of ≥ 19, well above the collapse level of $\approx$13 observed without repulsion.

Polyphonic diversity (unique bins ≥ 19, entropy ≥ 3.82 nats) is stable across the entire (b, c) grid. The only systematic trend is that C_{score} itself increases when the roughness penalty $|b|$ decreases or the interaction weight c increases, which is expected from the kernel definition; the emergent pitch structure, however, remains robust.

S7.4 Integration Test Terrain Control (Experiment 4)

To test whether the synergistic heredity$\times$hill-climbing interaction (main text, §5.1) depends on the coherent structure of the psychoacoustic landscape, we repeated all four integration-test conditions on a shuffled landscape, yielding a complete $2 \times 2 \times 2$ factorial (heredity/random $\times$ hill-climbing on/off $\times$ real/shuffled terrain; 8 conditions $\times$ 20 seeds $= 160$ runs).

On real terrain, the interaction contrast is large and significant (0.30, $p < 0.001$); on the shuffled landscape the contrast drops to 0.03 ($p = 0.19$), eliminating the synergy entirely. The full factorial also reveals that coherent terrain affects the non-hill conditions asymmetrically: heredity benefits substantially from real terrain even without hill-climbing (H real 0.185 vs. H shuffled 0.051, $p = 0.0002$), because consonance-gated metabolism rewards agents that happen to inherit consonant positions, an effect that vanishes when the landscape is spatially incoherent. Random respawn shows no such terrain dependence (R real 0.061 vs. R shuffled 0.059, $p = 0.70$), confirming that the metabolism-mediated terrain effect operates specifically through heritable pitch structure.

Table S1: Complete parameter listing for all experiments.

Parameter	**Symbol**	**Exp. 1**	**Exp. 2**	**Exp. 3**	**Exp. 4**
Anchor frequency	f_0	220 Hz	220 Hz	220 Hz	220 Hz
Population size	N	24	32	32	32
Seeds		20	20	20	20
Pitch range		± 1 oct	± 1 oct	± 1 oct	± 1 oct
Harmonicity / Roughness					
Harmonic order	N_H	16	16	16	16
Power-law exponent	ρ	0.4	0.4	0.4	0.4
Roughness saturation	k	1.0	1.0	1.0	1.0
Consonance core					
C_{field} coefficients	(a, b, c, d)	$(1, -1.35, 1, 0)$	$(1, -1.35, 1, 0)$	$(1, -1.35, 1, 0)$	$(1, -1.35, 1, 0)$
C_{density} coefficients[‡]	(a, b, c, d)	$(1, 0, -1, 0)$	—	—	—
Sigmoid (β, θ)		—	$(2, 0)$	—	$(2, 0)$
Pitch adaptation					
Step size	λ	0.15 st	—	—	—
Repulsion kernel	σ_r	0.06 st	—	—	0.06 st
Initial temperature	T_0	0.05	—	—	—
Annealing time constant	τ	30 steps	—	—	—
Duration		100 rounds	—	—	—
Phase switch		round 50	—	—	—
Burn-in discarded		10 rounds	—	—	—
Metabolism					
Basal cost	c_b	—	0.5/s	0.05/s[†]	0.5/s
Attack cost	c_a	—	0.02	—	0.02
Recharge coefficient	r_E	—	0.5/att	0.1/s[†]	0.5/att
Energy cap	E_{cap}	—	1.0	1.0	1.0
Time step	Δt	—	512/48k s	0.02 s	512/48k s
Termination		—	200 deaths	40 s	4000 deaths
Temporal coupling					
Coupling strength	K	—	—	3.0	—
External kick	f_{kick}	—	—	2 Hz	—
Intrinsic frequency	ω_0	—	—	$1.8 \cdot 2\pi$	—
Frequency jitter		—	—	$\pm 2\%$	—
Heredity					
Mutation σ	σ_m	—	—	—	$0.003\ \log_2$
Parent selection		—	—	—	uniform
JI cross-validation					
Proximity kernel	α	—	—	—	$4.0\ \text{st}^{-2}$
Ratio limit	p, q	—	—	—	≤ 8
Ratio weight		—	—	—	$1/(pq)$

[†] Experiment 3 uses a simplified continuous energy model (see S1.3).

[‡] Visualization only (Figure 1D); not used in experiment algorithms.

Table S2: Full parameter sweep for Experiment 4 (hereditary adaptation). Each row reports one parameter setting; "Sep." is the mean ΔC_{score}(heredity) $-$ mean ΔC_{score}(random). Welch's t-test compares the two arms ($n = 20$ seeds each). $\star$ marks the setting reported in the main text.

σ	Deaths	N	$\overline{\Delta C}_{\text{her}}$	$\overline{\Delta C}_{\text{rnd}}$	Sep.	t	p
0.001	2000	32	0.074 ± 0.150	0.004 ± 0.035	0.071	2.06	0.052
0.001	3000	32	0.117 ± 0.174	0.007 ± 0.025	0.111	2.81	0.011*
0.002	2000	32	0.067 ± 0.119	0.004 ± 0.035	0.063	2.29	0.032*
0.002	3000	32	0.039 ± 0.093	0.007 ± 0.025	0.032	1.49	0.151
0.003	2000	16	0.035 ± 0.101	-0.004 ± 0.036	0.039	1.62	0.118
0.003	2000	24	0.089 ± 0.155	0.004 ± 0.029	0.085	2.41	0.025*
0.003	2000	32	0.096 ± 0.138	0.004 ± 0.035	0.093	2.91	0.008*
0.003	2000	48	0.061 ± 0.141	0.004 ± 0.021	0.058	1.80	0.087
0.003	3000	32	0.118 ± 0.160	0.007 ± 0.025	0.111	3.07	0.006*
$\star$0.003	4000	32	0.121 ± 0.177	-0.003 ± 0.027	0.124	3.10	0.006*
0.003	5000	32	0.113 ± 0.167	-0.003 ± 0.025	0.116	3.07	0.006*
0.005	2000	32	0.078 ± 0.135	0.004 ± 0.035	0.075	2.40	0.026*
0.005	3000	32	0.072 ± 0.119	0.007 ± 0.025	0.065	2.39	0.027*
0.01	2000	32	0.058 ± 0.102	0.004 ± 0.035	0.054	2.25	0.034*
0.02	2000	32	0.106 ± 0.135	0.004 ± 0.035	0.102	3.26	0.004*

*Significant at $p < 0.05$. $\star$ = reported setting in the main text ($\sigma = 0.003$, 4,000 deaths, $N = 32$).

Table S3: Curriculum vs. consonance-only control (Experiment 1, baseline condition, $n = 20$ seeds). All differences are significant at $p < 0.001$ (Welch's t-test).

Metric	**Curriculum**	**Cons.-only**	t
C_{score} (post)	0.474 ± 0.040	0.538 ± 0.036	5.3
Unique pitch bins	19.2 ± 1.3	14.8 ± 1.8	-8.8
NN distance (st)	0.574 ± 0.048	0.353 ± 0.095	-9.2
Interval entropy	3.93 ± 0.16	3.53 ± 0.32	-5.0

Table S4: Baseline vs. Shuffled landscape (Experiment 1).

Metric	**Baseline**	**Shuffled**	t
Unique pitch bins	19.2 ± 1.3	19.5 ± 1.2	0.6
NN distance (st)	0.574 ± 0.048	0.568 ± 0.079	-0.3
Interval entropy	3.93 ± 0.16	4.65 ± 0.10	17.2^{***}
C_{score} (ref)	0.474 ± 0.040	0.388 ± 0.031	-7.6^{***}

Table S5: Baseline vs. H/R-misaligned conditions (Experiment 1).

Condition	**Unique bins**	**NN dist. (st)**	**Entropy**
Baseline	19.2 ± 1.3	0.574 ± 0.048	3.93 ± 0.16
$\Delta = 137\,¢$	18.8 ± 1.3	0.547 ± 0.099	3.87 ± 0.17
$\Delta = 271\,¢$	19.0 ± 1.5	0.581 ± 0.093	3.92 ± 0.15
$\Delta = 389\,¢$	19.3 ± 1.9	0.595 ± 0.093	3.85 ± 0.26
$\Delta = 523\,¢$	19.1 ± 1.5	0.590 ± 0.080	3.91 ± 0.19
Misaligned (avg)	19.0 ± 1.6	0.578 ± 0.094	3.89 ± 0.20

Table S6: Coefficient sweep: polyphony metrics across 16 (b, c) settings.

b	c	Unique bins	NN (st)	Entropy	C_{score}
-2.00	0.00	20.0 ± 1.7	0.620 ± 0.081	3.99 ± 0.21	0.356 ± 0.046
-2.00	0.50	19.9 ± 1.3	0.593 ± 0.069	4.00 ± 0.21	0.391 ± 0.046
-2.00	1.00	19.2 ± 1.8	0.561 ± 0.100	3.91 ± 0.20	0.413 ± 0.038
-2.00	1.50	19.8 ± 1.8	0.586 ± 0.085	3.90 ± 0.21	0.445 ± 0.045
-1.35	0.00	19.7 ± 1.5	0.590 ± 0.056	3.85 ± 0.24	0.436 ± 0.034
-1.35	0.50	19.5 ± 1.2	0.588 ± 0.087	3.92 ± 0.24	0.449 ± 0.044
-1.35	1.00	19.2 ± 1.3	0.574 ± 0.048	3.93 ± 0.16	0.474 ± 0.040
-1.35	1.50	19.5 ± 1.5	0.573 ± 0.081	3.95 ± 0.22	0.491 ± 0.039
-0.70	0.00	20.2 ± 1.4	0.630 ± 0.095	3.89 ± 0.22	0.483 ± 0.034
-0.70	0.50	19.9 ± 1.2	0.617 ± 0.089	3.87 ± 0.21	0.511 ± 0.039
-0.70	1.00	19.5 ± 1.3	0.578 ± 0.082	3.89 ± 0.21	0.538 ± 0.049
-0.70	1.50	19.4 ± 1.0	0.592 ± 0.072	3.93 ± 0.17	0.558 ± 0.045
0.00	0.00	19.0 ± 1.4	0.566 ± 0.092	3.89 ± 0.19	0.547 ± 0.039
0.00	0.50	19.0 ± 1.6	0.583 ± 0.104	3.84 ± 0.28	0.556 ± 0.041
0.00	1.00	19.5 ± 1.1	0.596 ± 0.083	3.92 ± 0.15	0.593 ± 0.053
0.00	1.50	19.2 ± 1.4	0.592 ± 0.086	3.82 ± 0.20	0.619 ± 0.042

Table S7: Full $2\times2\times2$ integration terrain control (20 seeds each).

Condition	Real terrain	Shuffled	Difference
H (heredity, no hill)	0.185 ± 0.076	0.051 ± 0.006	0.134^{***}
HH (heredity + hill)	0.530 ± 0.068	0.082 ± 0.049	0.449^{***}
R (random, no hill)	0.061 ± 0.008	0.059 ± 0.008	0.003
RH (random + hill)	0.107 ± 0.013	0.056 ± 0.006	0.051^{***}

Interaction contrast (real): 0.30 [0.20, 0.39], $p < 0.001$
Interaction contrast (shuffled): 0.03 [-0.02, 0.08], $p = 0.19$